

PL/SQL Programs 1 to 15

Simple Human-Written Practical Programs

Before running the programs, use: SET SERVEROUTPUT ON;

1. Print "Welcome To PL/SQL Block"

```
DECLARE
BEGIN
    DBMS_OUTPUT.PUT_LINE('Welcome To PL/SQL Block');
END;
/
```

2. Program to Count Square

```
DECLARE
    n NUMBER;
    sq NUMBER;
BEGIN
    n := &n;

    sq := n * n;

    DBMS_OUTPUT.PUT_LINE('Square = ' || sq);
END;
/
```

3. Program to Find Simple Interest

```
DECLARE
    p NUMBER;
    r NUMBER;
    t NUMBER;
    si NUMBER;
BEGIN
    p := &p;
    r := &r;
    t := &t;

    si := (p * r * t) / 100;

    DBMS_OUTPUT.PUT_LINE('Simple Interest = ' || si);
END;
/
```

4. Program for Global and Local Variable

```
DECLARE
    a NUMBER := 10;

    PROCEDURE show_value
    IS
        b NUMBER := 20;
    BEGIN
        DBMS_OUTPUT.PUT_LINE('Global Variable = ' || a);
        DBMS_OUTPUT.PUT_LINE('Local Variable = ' || b);
    END;
BEGIN
    show_value;
END;
/
```

5. Program to Declare Constant Variable

```
DECLARE
    pi CONSTANT NUMBER := 3.14;
    r NUMBER;
    area NUMBER;
BEGIN
    r := &r;

    area := pi * r * r;

    DBMS_OUTPUT.PUT_LINE('Area of Circle = ' || area);
END;
/
```

6. Ticket Age Wise Using IF-THEN-ELSIF-ELSE

```
DECLARE
    age NUMBER;
    ticket NUMBER;
BEGIN
    age := &age;

    IF age < 5 THEN
        ticket := 0;
    ELSIF age <= 12 THEN
        ticket := 50;
    ELSIF age <= 60 THEN
        ticket := 100;
    ELSE
        ticket := 70;
    END IF;

    DBMS_OUTPUT.PUT_LINE('Ticket Price = ' || ticket);
END;
/
```

7. Find Maximum and Minimum from Three Numbers

```
DECLARE
    a NUMBER;
    b NUMBER;
    c NUMBER;
    max NUMBER;
    min NUMBER;
BEGIN
    a := &a;
    b := &b;
    c := &c;

    IF a > b AND a > c THEN
        max := a;
    ELSIF b > c THEN
        max := b;
    ELSE
        max := c;
    END IF;

    IF a < b AND a < c THEN
        min := a;
    ELSIF b < c THEN
        min := b;
    ELSE
        min := c;
    END IF;

    DBMS_OUTPUT.PUT_LINE('Maximum = ' || max);
    DBMS_OUTPUT.PUT_LINE('Minimum = ' || min);
END;
/
```

8. Roll No, Marks, Total, Percentage, Result and Grade

```
DECLARE
    rollno NUMBER;
    m1 NUMBER;
    m2 NUMBER;
    m3 NUMBER;
    total NUMBER;
    per NUMBER;
    result VARCHAR2(10);
```

```

        grade VARCHAR2(10);
BEGIN
    rollno := &rollno;
    m1 := &m1;
    m2 := &m2;
    m3 := &m3;

    total := m1 + m2 + m3;
    per := total / 3;

    IF m1 < 35 OR m2 < 35 OR m3 < 35 THEN
        result := 'Fail';
        grade := 'F';
    ELSE
        result := 'Pass';

        IF per >= 80 THEN
            grade := 'A';
        ELSIF per >= 60 THEN
            grade := 'B';
        ELSIF per >= 50 THEN
            grade := 'C';
        ELSE
            grade := 'D';
        END IF;
    END IF;

    DBMS_OUTPUT.PUT_LINE('Roll No = ' || rollno);
    DBMS_OUTPUT.PUT_LINE('Total = ' || total);
    DBMS_OUTPUT.PUT_LINE('Percentage = ' || per);
    DBMS_OUTPUT.PUT_LINE('Result = ' || result);
    DBMS_OUTPUT.PUT_LINE('Grade = ' || grade);
END;
/

```

9. Program for PL/SQL CASE Statement

```

DECLARE
    n NUMBER;
BEGIN
    n := &n;

    CASE n
        WHEN 1 THEN
            DBMS_OUTPUT.PUT_LINE('Monday');
        WHEN 2 THEN
            DBMS_OUTPUT.PUT_LINE('Tuesday');
        WHEN 3 THEN
            DBMS_OUTPUT.PUT_LINE('Wednesday');
        WHEN 4 THEN
            DBMS_OUTPUT.PUT_LINE('Thursday');
        WHEN 5 THEN
            DBMS_OUTPUT.PUT_LINE('Friday');
        WHEN 6 THEN
            DBMS_OUTPUT.PUT_LINE('Saturday');
        WHEN 7 THEN
            DBMS_OUTPUT.PUT_LINE('Sunday');
        ELSE
            DBMS_OUTPUT.PUT_LINE('Invalid Number');
    END CASE;
END;
/

```

10. Print First 10 Numbers Using Basic LOOP and EXIT

```

DECLARE

```

```
        i NUMBER := 1;
BEGIN
    LOOP
        DBMS_OUTPUT.PUT_LINE(i);

        i := i + 1;

        EXIT WHEN i > 10;
    END LOOP;
END;
/
```

11. Print First 10 Numbers Using WHILE LOOP

```
DECLARE
    i NUMBER := 1;
BEGIN
    WHILE i <= 10 LOOP
        DBMS_OUTPUT.PUT_LINE(i);

        i := i + 1;
    END LOOP;
END;
/
```

12. Print First 10 Numbers Using FOR LOOP

```
BEGIN
    FOR i IN 1..10 LOOP
        DBMS_OUTPUT.PUT_LINE(i);
    END LOOP;
END;
/
```

13. Print 1 to 10 Numbers in Reverse Using FOR LOOP

```
BEGIN
    FOR i IN REVERSE 1..10 LOOP
        DBMS_OUTPUT.PUT_LINE(i);
    END LOOP;
END;
/
```

14. Print First 10 Prime Numbers Using WHILE LOOP

```
DECLARE
    n NUMBER := 2;
    count_prime NUMBER := 0;
    i NUMBER;
    count_div NUMBER;
BEGIN
    WHILE count_prime < 10 LOOP

        i := 1;
        count_div := 0;

        WHILE i <= n LOOP
            IF MOD(n, i) = 0 THEN
                count_div := count_div + 1;
            END IF;

            i := i + 1;
        END LOOP;

        IF count_div = 2 THEN
            DBMS_OUTPUT.PUT_LINE(n);
            count_prime := count_prime + 1;
        END IF;

        n := n + 1;
    END LOOP;
END;
/
```

15. Accept a String and Display Reverse

```
DECLARE
```

```
    str VARCHAR2(100);
    rev VARCHAR2(100) := '';
BEGIN
    str := '&str';

    FOR i IN REVERSE 1..LENGTH(str) LOOP
        rev := rev || SUBSTR(str, i, 1);
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('Reverse = ' || rev);
END;
/
```